Postulates & Diagrams — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: What is a postulate?*

Key Idea

A **postulate** is a rule we accept as true **without proof**. Everything else in geometry is built on postulates.

☞ *Video: Point, line and plane postulates*

Key Idea

Through any **two points** there is exactly one **line**.



Through any three **noncollinear** points there is exactly one **plane**.

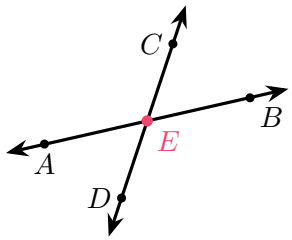


Two lines meet in exactly one **point**.

Two planes meet in exactly one **line**.

☞ *Video: Reading a diagram*

Watch Me — Worked Example



A, E, B are collinear — **can** assume

the two lines meet at E — **can** assume

$\angle AEC = 90°$ — **cannot** — no **right-angle mark**

$AE = EB$ — **cannot** — no **tick marks**

☞ *Video: Marks tell you sizes*

Key Idea



right angle



equal lengths



parallel

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☞ *Video: Watch out*

Common Mistake

"It looks like it" is __not__ a reason. Only what is __drawn__ (position) or __marked__ (size) counts.

☞ *Video: Your turn*

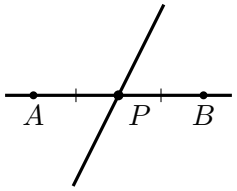
Your Turn

Which can you assume?

(a) A, P, B are collinear __yes__ — drawn on one line

(b) $PA = PB$ __yes__ — __tick marks__

(c) the lines are perpendicular __no__ — no right-angle mark



Notice

Remember: Two points $\rightarrow$ one line; three noncollinear points $\rightarrow$ one plane. Two lines meet in a point; two planes meet in a line. Assume what is **drawn**, not sizes — unless they are **marked**.

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